



How AI meets Healthcare



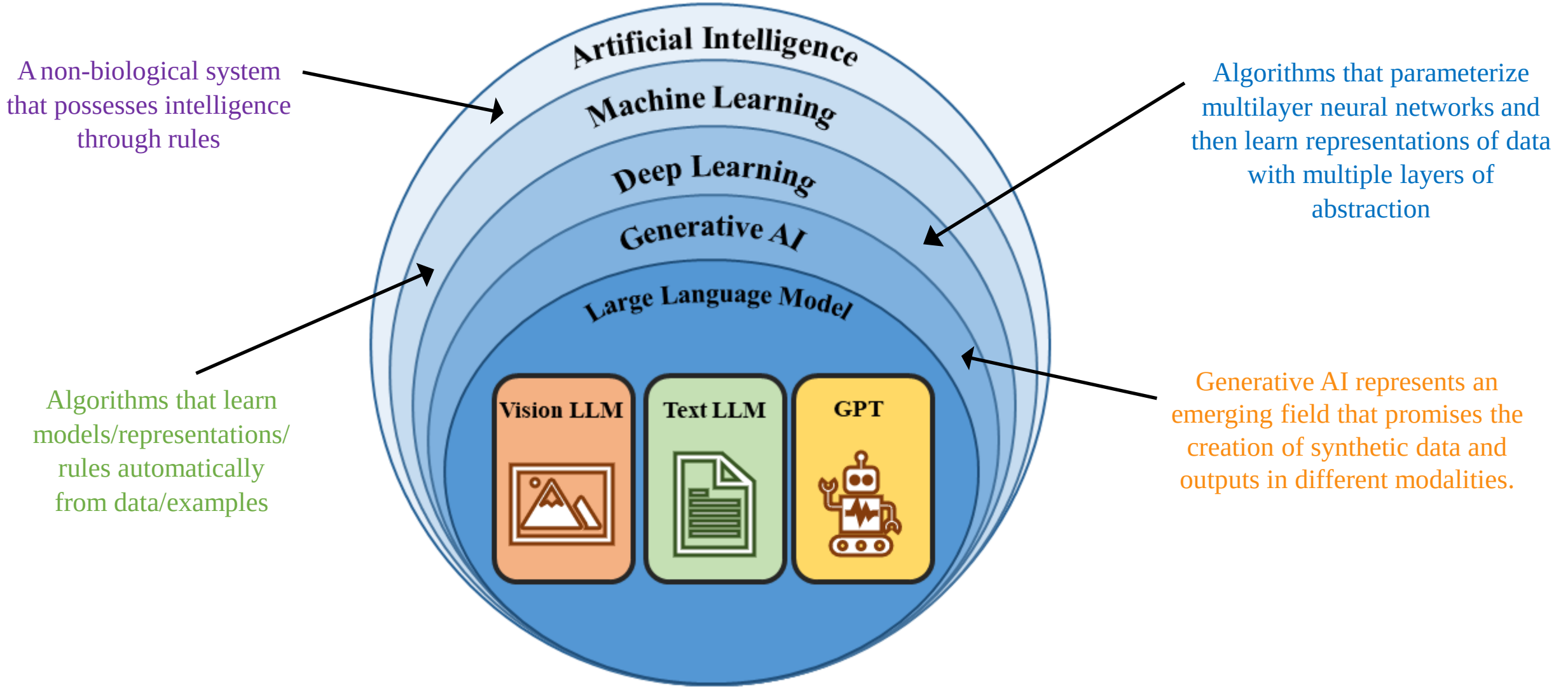
GLOBAL
CARE
Hospital



Dr. Tanujit Chakraborty
Sorbonne University Abu Dhabi
Sorbonne Center for AI, Paris
Duke-NUS Medical School, Singapore



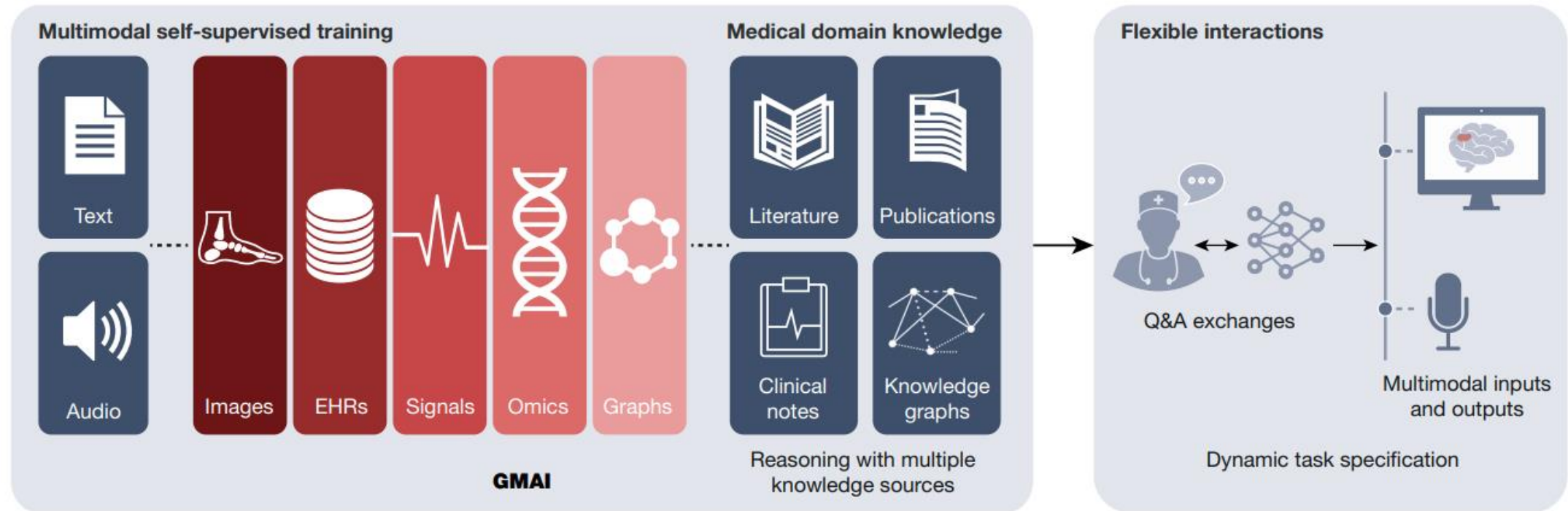
AI-ML Tools



Large language models (LLMs) are AI models applied to various tasks in health care, ranging from answering medical examination questions to generating clinical reports.

AI in Health

a



b

Applications



Chatbots for patients



Interactive note-taking



Augmented procedures

...



Grounded radiology reports



Text-to-protein generation



Bedside decision support

Regulations: Application approval; validation; audits; community-based challenges; analyses of biases, fairness and diversity

Health Data Science

AI in Health Machine Learning & Forecasting

mHealth

Data

- Drink less study
- Micro-Randomized Trials (MRTs)
- Dynamic treatment regime



Epidemic TS

centric

- Time Series Forecasting
 - Risk assessment
 - Infectious Disease modeling



Interpretable ML

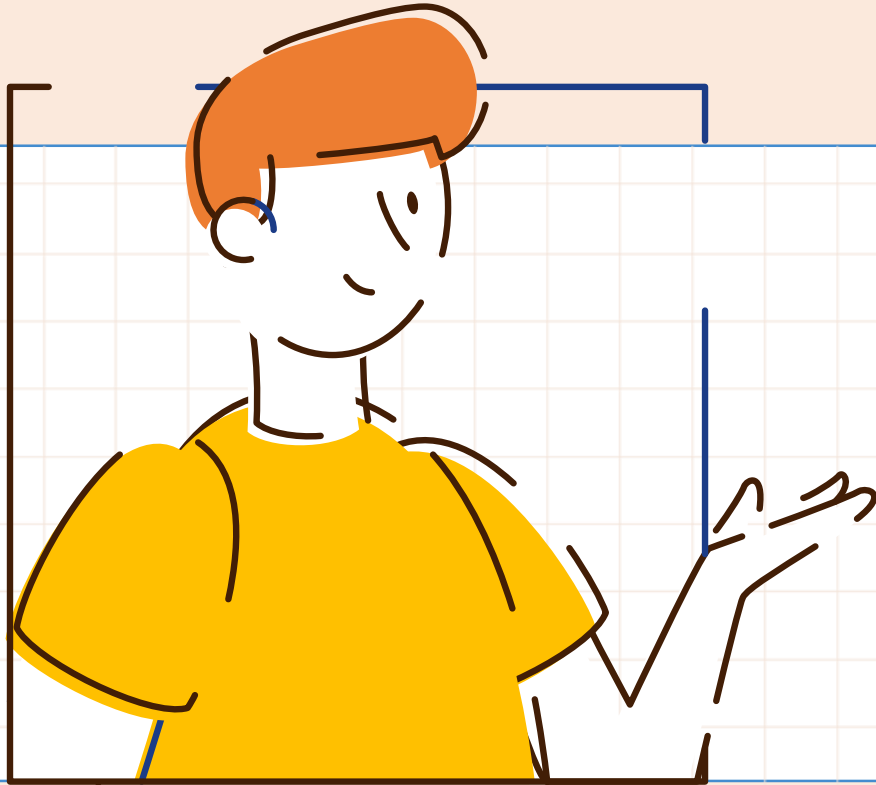
view

- Survival Analysis on EHR
- Causality Analysis
- Explainable and Generative AI



Mobile Health

PART 01



Submitted to the Annals of Applied Statistics

THOMPSON SAMPLING FOR ZERO-INFLATED COUNT OUTCOMES WITH AN APPLICATION TO THE DRINK LESS MOBILE HEALTH STUDY

BY XUEQING LIU^{1,a}, NINA DELIU^{2,3,c} TANUJIT CHAKRABORTY^{4,f}
LAUREN BELL^{3,5,c} AND BIBHAS CHAKRABORTY^{1,6,7,b}



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Mobile Health

Mobile health (mHealth) refers to the use of mobile technologies for managing one's health and wellness

Mobile/digital interventions can potentially reduce health disparity, and thus appealing from a global health perspective!

Health Applications: Alcohol use disorders, Mindfulness and emotional regulation in expectant couples, Physical activity management



Drink less Study

- *Drink Less* helps people cut back on hazardous alcohol use
- **Engagement** with *Drink Less* is crucial for the distal effect of reducing alcohol consumption

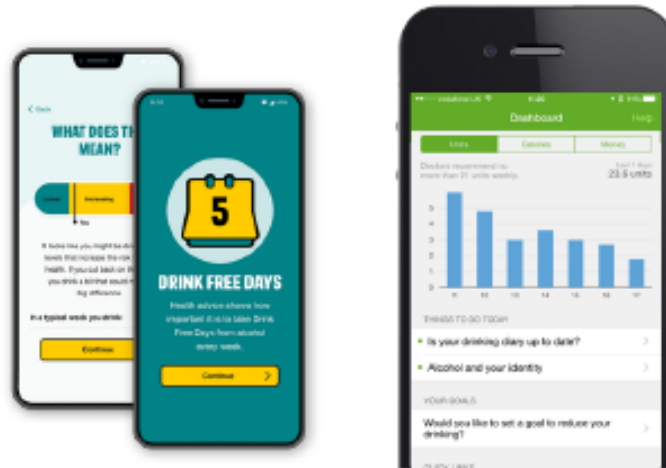
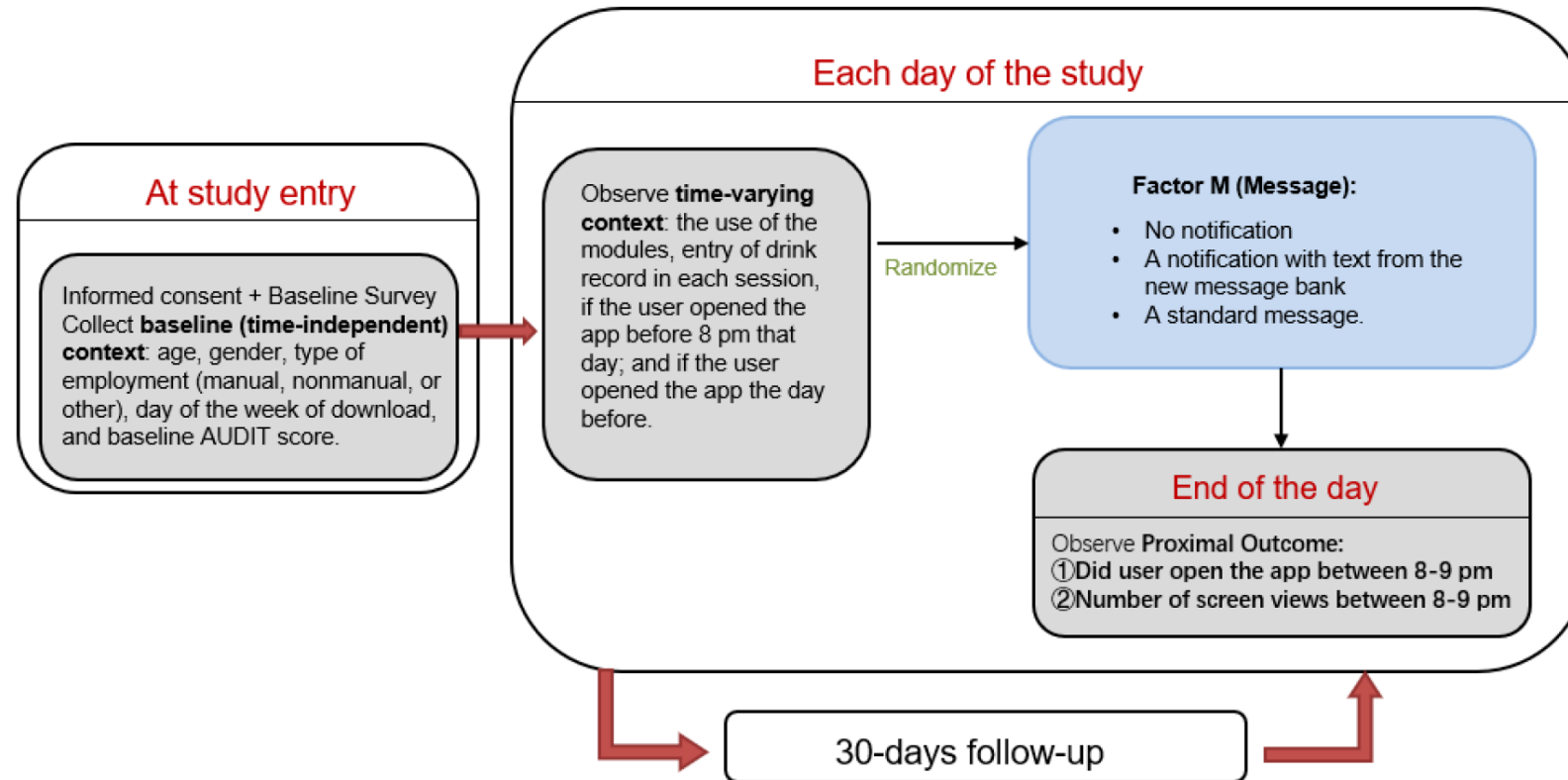


Figure: A behavior change app named *Drink Less*

- Count-valued proximal outcome: number of screen views in the subsequent hour of intervention
- Contextual variables: age, gender, AUDIT score, day since download, occupation type,...

Drink less Study



- A micro-randomized trial to promote engagement with the Drink Less app among people who have hazardous alcohol consumption
- Which notification can influence near-term engagement?
- In which context should the smart mobile phone send the user a notification to use the app?

Drink Less

Goal: Maximize the cumulative proximal outcome

Algorithms

$$\max \sum_{t=1}^T Y_t$$



Notifications

Interventions / Actions

A_t

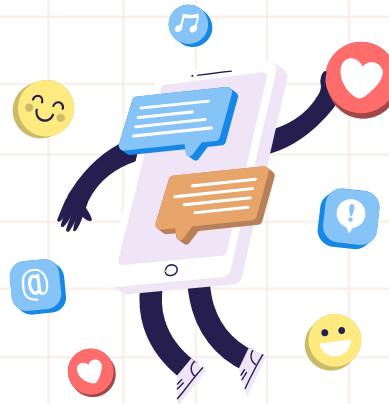
At risk
Not at risk
Context

X_t



People with alcohol use disorder

User



Engage with Drink Less app:

Number of screen views

Proximal outcome

Y_t



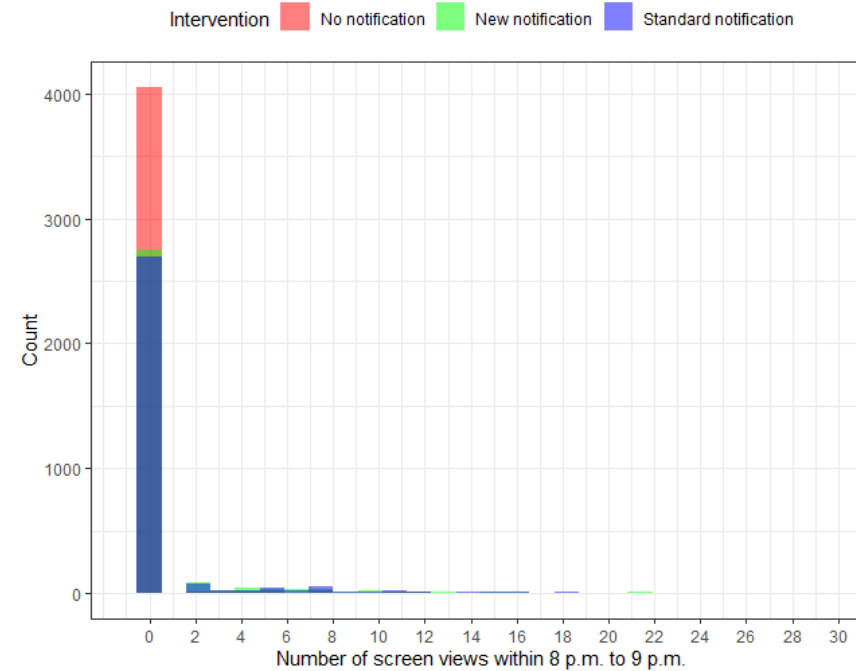
Reduce alcohol consumption

Distal outcome

Drink less

Engage with Drink Less app:

Number of screen views following each intervention



Data: At each decision time t , the mobile phone collects tailoring variable X_t and chooses an intervention A_t , and then obtains a proximal outcome Y_t

Background Literature

- **Goal:** Continuously learn the best ways to develop and push interventions to improve their weekly wellness by maximizing proximal outcomes.
- **Evaluation Metric: Regret** (difference between the expected outcome of a candidate learning algorithm and the optimal expected outcome by an oracle)

Approaches:

- Learning to take better actions (interventions) in an environment by interacting with it has been studied in the field of **Reinforcement Learning (RL)** with classical applications in **robotics, automated flights, games, etc.**
- Specifically, short-horizon RL algorithms, like **contextual multi-armed bandit (MAB)** algorithms are suitable for mHealth interventions (e.g., text-messages) as well as **online ads**.
- In particular, we focus on an algorithm called **Linear Thompson Sampling** (Thompson, 1933; Agarwal and Goyal, 2013), operationalized by a Bayesian regression (**a multi-step reward-based algorithm**).
- **Count model:** Poisson regression, negative binomial regression, zero-inflated models...

Drink less

What have we done?

- We propose a strategy, TS-Count, for optimizing intervention delivery in mHealth applications with count-valued proximal outcomes
- We provide insight into the effect and handling of overdispersion/zero-inflation in an online setting
- Theoretical analysis of TS-ZIP and TS-ZINB

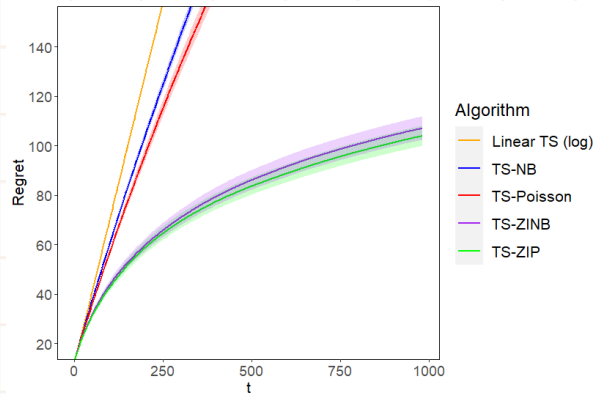
Future directions:

- Extending to multi-user setting that enables the borrowing of information between users and the accommodation of heterogeneity

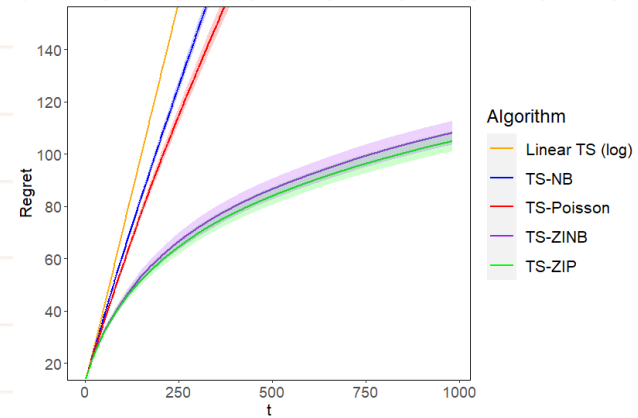


Results

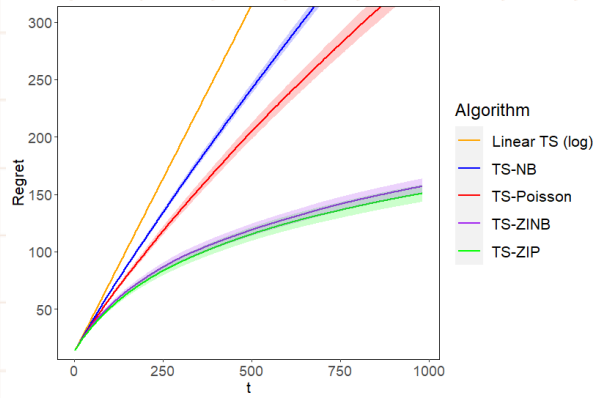
Outcome generated from a zero-inflated Poisson distribution



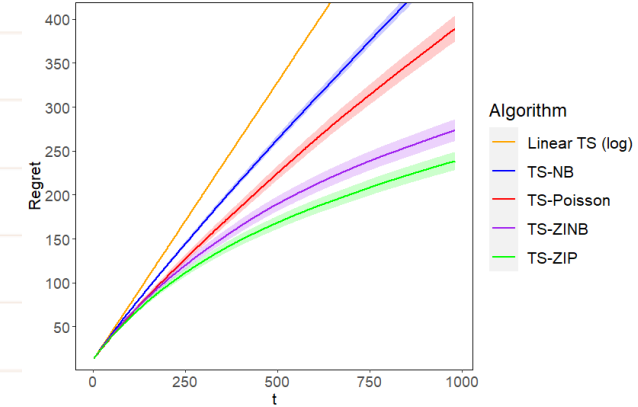
No overdispersion



Low overdispersion



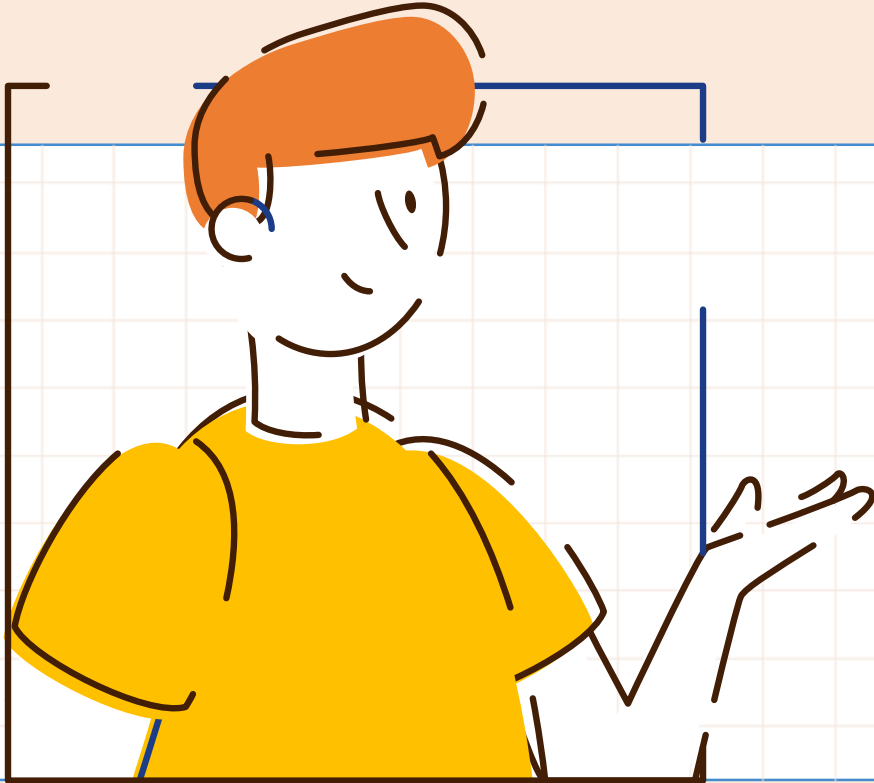
Moderate overdispersion



High overdispersion

Epidemic Time Series

PART 02



Neural Networks
Volume 165, August 2023, Pages 185-212



Epicasting: An Ensemble Wavelet Neural Network for forecasting epidemics

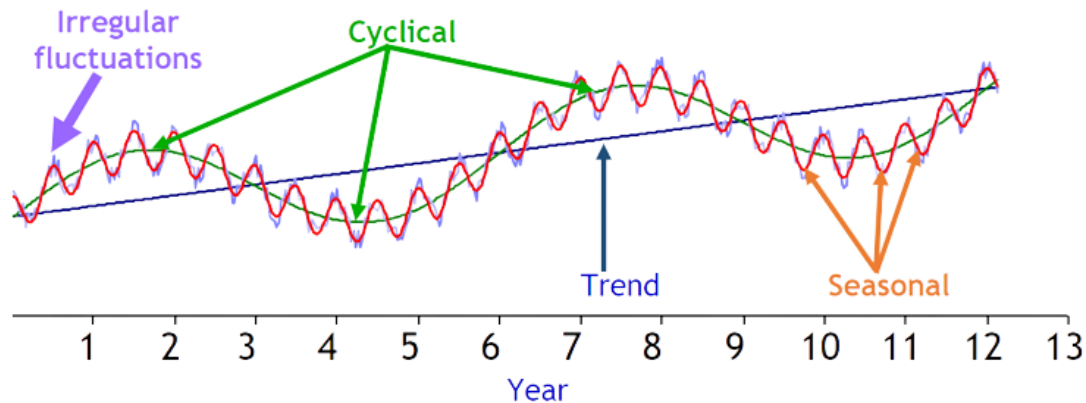
Madhurima Panja^{a,1}, Tanujit Chakraborty^{b, a, c, 1}  , Uttam Kumar^a, Nan Liu^d



DukeNUS
Medical School

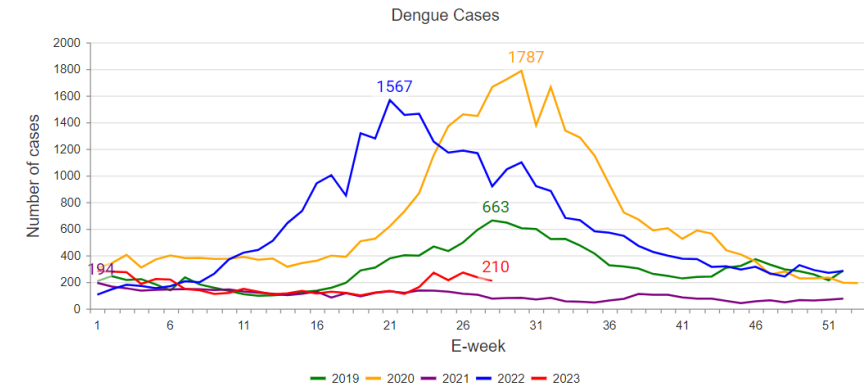
What is time series?

Time series is a set of observations, each one being recorded at a specific time.

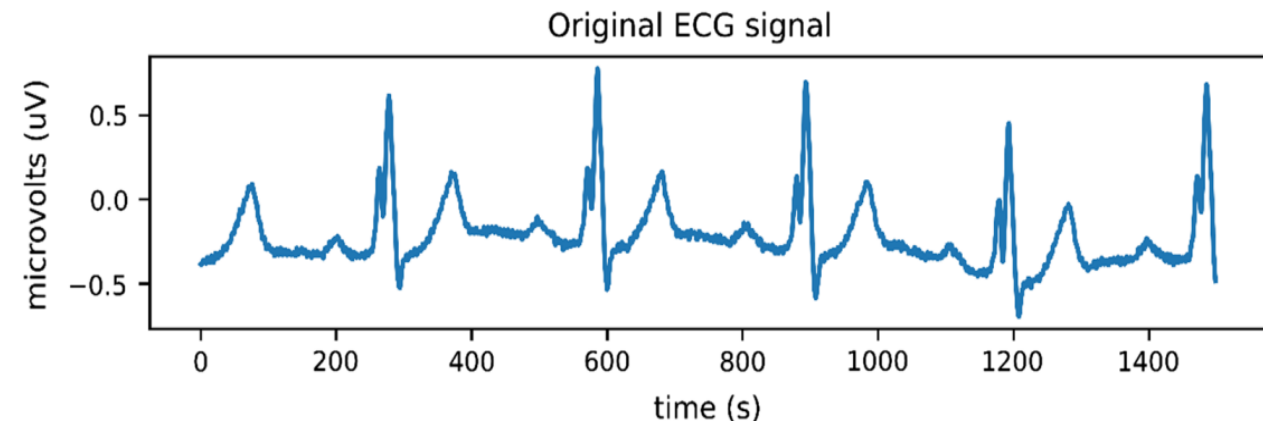


Forecasting is estimating how the sequence of observations will continue into the future.

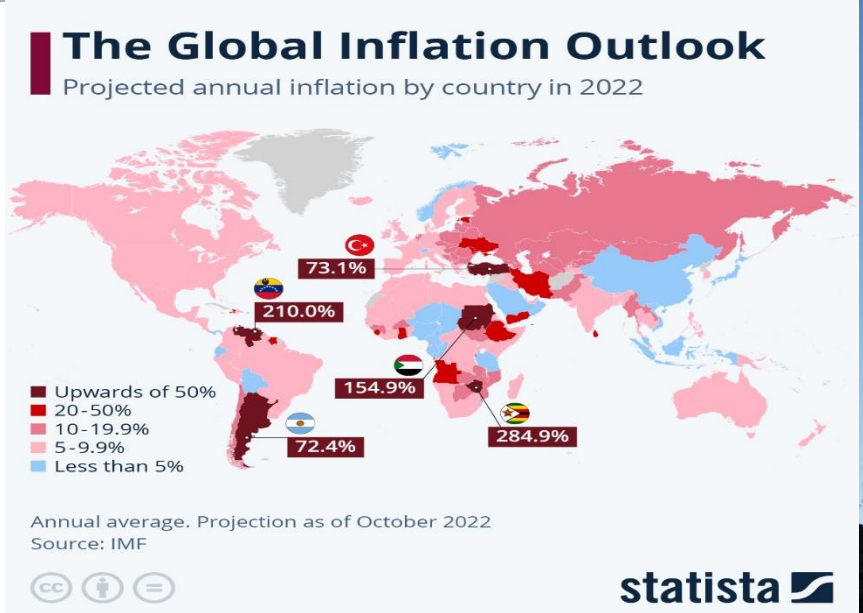
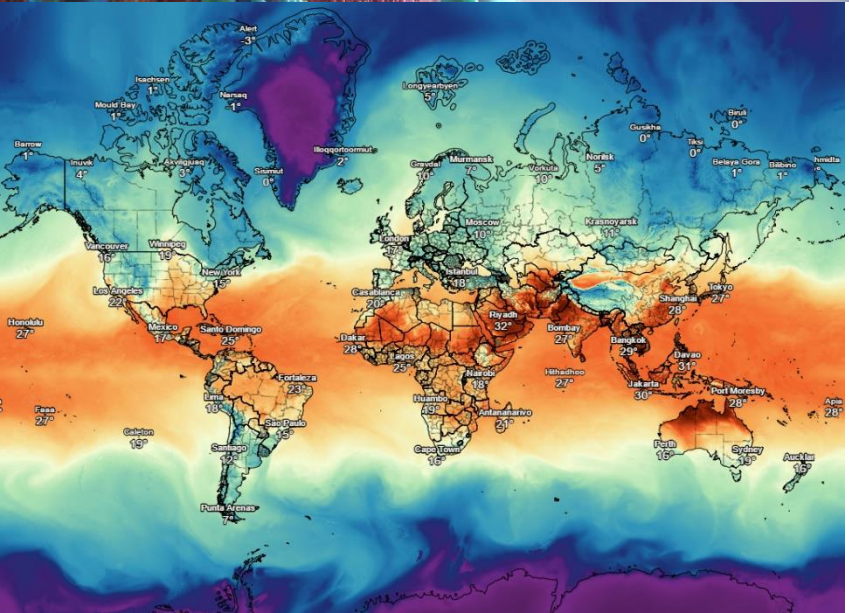
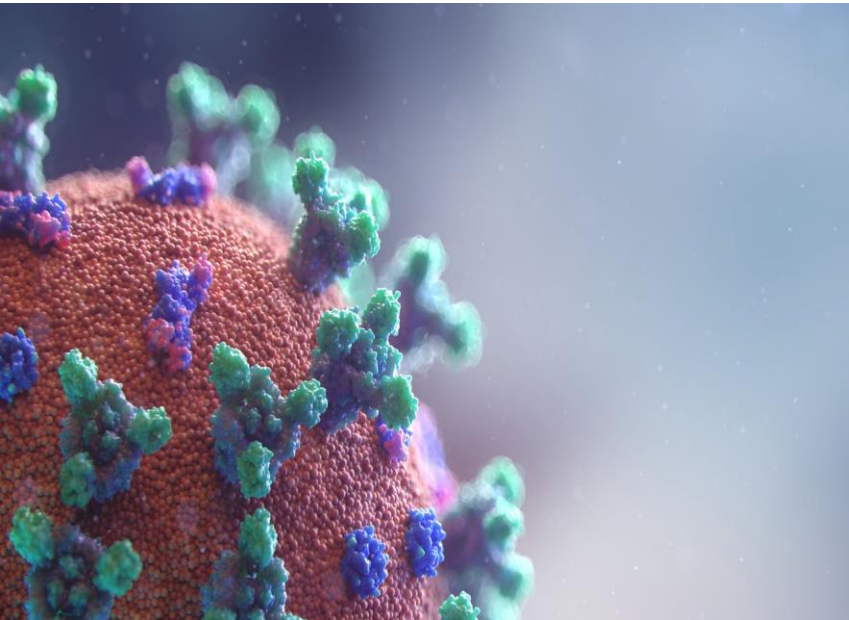
Discrete time series is one in which the set of time points at which observations are made is a discrete set (e.g., weekly dengue incidence cases in SG)



Continuous time series are obtained when observations are made continuously over some time intervals (e.g., ECG graph)



What can we forecast?



Process of disease forecasting

nature communications

Technology to advance infectious disease forecasting for outbreak management

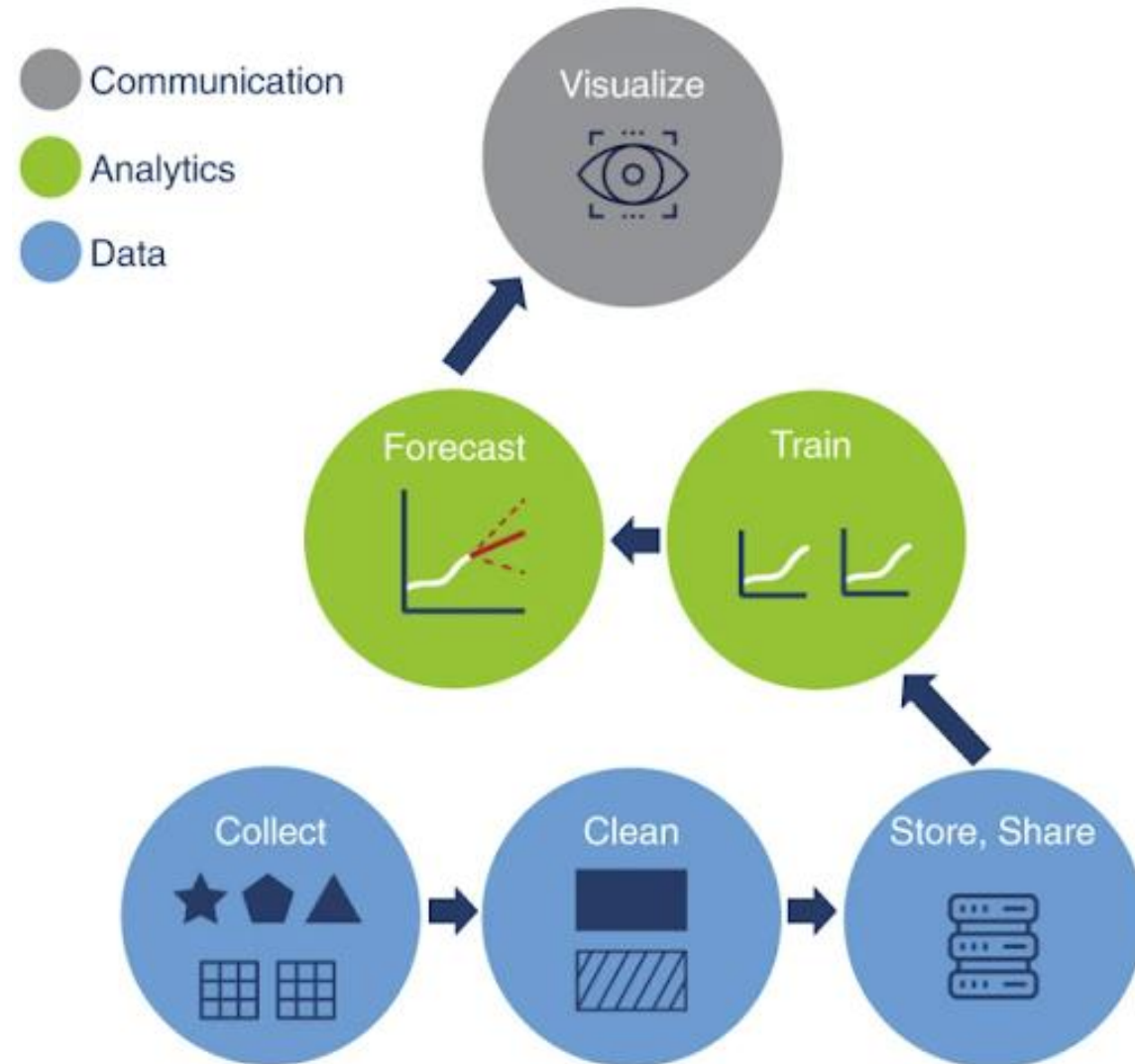
[Dylan B. George](#) , [Wendy Taylor](#), [Jeffrey Shaman](#), [Caitlin Rivers](#), [Brooke Paul](#), [Tara O'Toole](#), [Michael A. Johansson](#), [Lynette Hirschman](#), [Matthew Biggerstaff](#), [Jason Asher](#) & [Nicholas G. Reich](#)

[Nature Communications](#) **10**, Article number: 3932 (2019) | [Cite this article](#)

10k Accesses | 24 Citations | 37 Altmetric | [Metrics](#)

Forecasting is beginning to be integrated into decision-making processes for infectious disease outbreak response. We discuss how technologies could accelerate the adoption of forecasting among public health practitioners, improve epidemic management, save lives, and reduce the economic impact of outbreaks.

'Data gaps undermine our ability to target resources, develop policies and track accountability. Without good data, we're flying blind. If you can't see it, you can't solve it.' - [Kofi Annan \(Nature, 2018\)](#)



Benefits of epicasting

- To Governments:
 - Timing and focus of communications (e.g., vaccination campaigns)
 - Antiviral policy
 - Mosquito control, door-to-door campaigns
- To Health Care Providers:
 - Staffing, vacations
 - Elective surgery
 - Equipment pre-positioning
- To Individuals:
 - Protect our families (old and comorbidity)

Vision:

1. Building an epidemic forecasting model to handle the data irregularities
2. State-of-the-art performance
3. Understanding the theoretical and computational aspect
4. Building software for public use (similar to weather forecasts)
5. Making an impact in healthcare

Proposed EWNNet Model Architecture

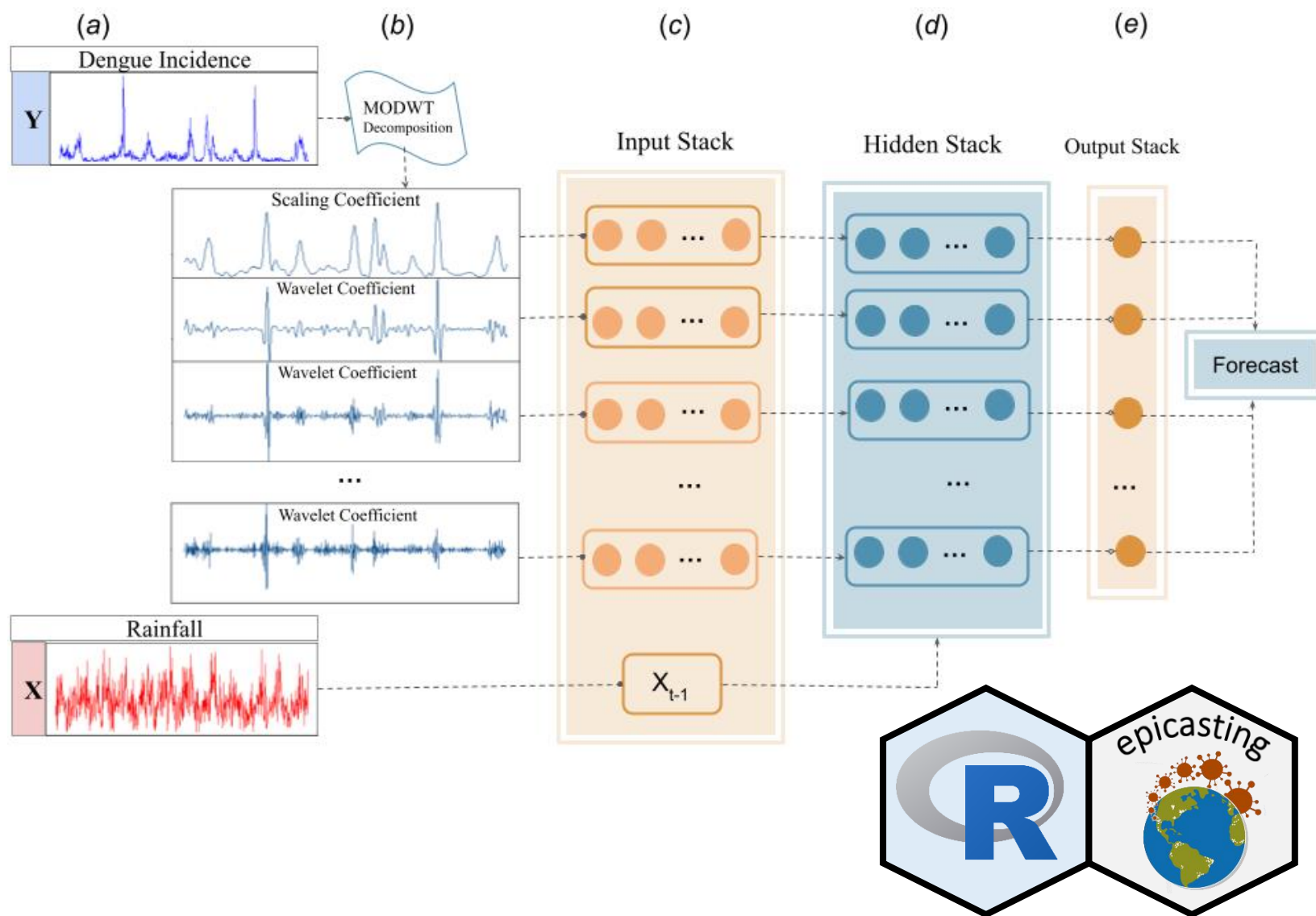


Figure: The proposed EWNNet workflow:

- To predict dengue incidence cases, we provide a weekly time series of dengue cases (Y_t) and rainfall (X_t) in the training period;
- We perform a MODWT based MRA transformation on Y and generate multiple series of details and smooth coefficients;
- We begin to train local auto-regressive neural networks to individually model the transformed series along with rainfall dataset in the input stack;
- Each of the neural networks is trained with a single hidden layer having a pre-specified number of nodes inside the hidden stack;
- The output stack comprises of one-step ahead forecast generated by individual neural networks. These predictions are combined to generate the final out-of-sample forecast.

Actual vs Forecast Visualization

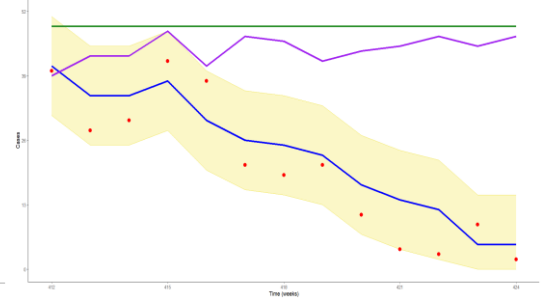
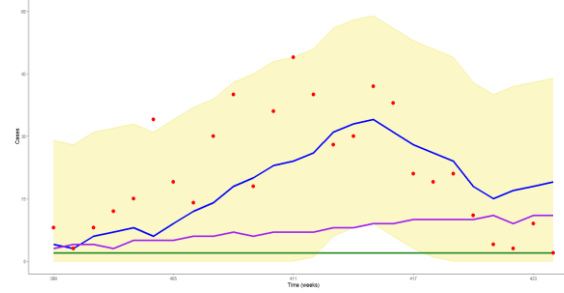
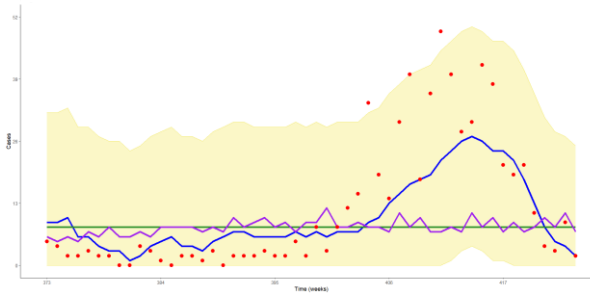
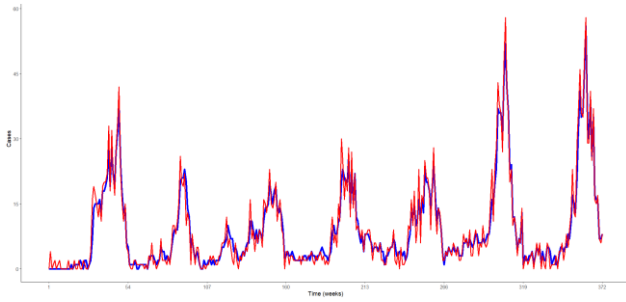
Training data and fitted values

Long-term forecast

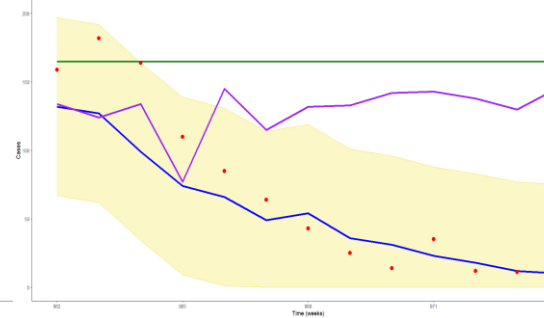
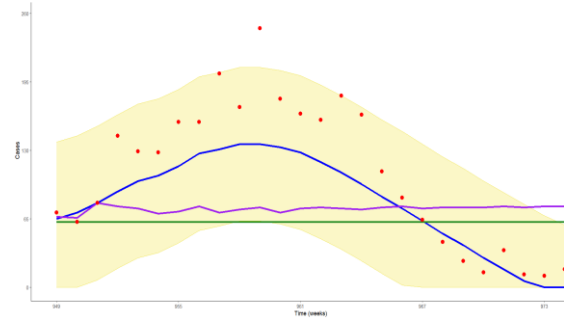
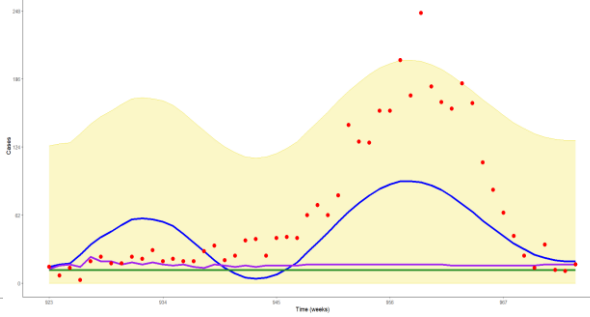
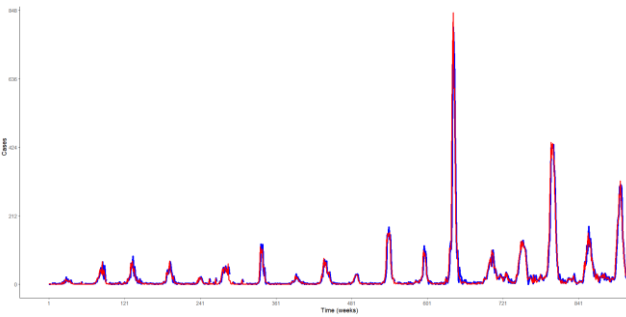
Medium-term forecast

Short-term forecast

Ahmedabad
Dengue



Australia Flu



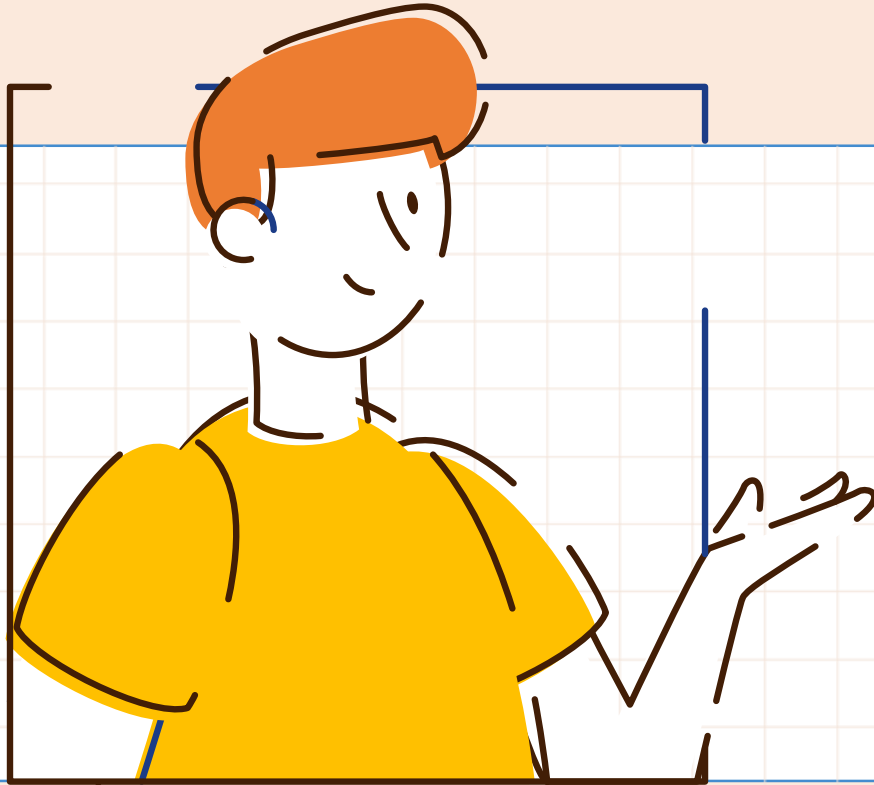
Data are collected from:



DENGUE FORECASTING
NATIONAL OCEANIC AND ATMOSPHERIC ADMINISTRATION

Explainable AI

PART 03



Submitted for Publication

Comparative survival analysis for predicting mortality after emergency admission

Ziwen Wang^{1#}, Jin Wee Lee^{1#}, Tanujit Chakraborty², Yilin Ning¹, Mingxuan Liu¹, Feng Xie^{1,3}, Marcus

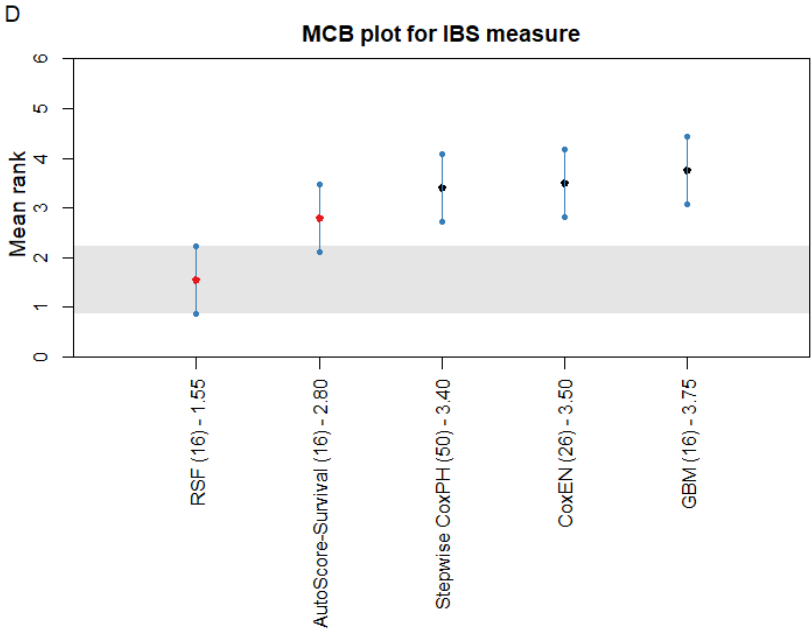
Eng Hock Ong^{3,4}, Nan Liu^{1,3,5*}

Interpretable ML for Survival Analysis

- A retrospective cohort study was conducted on patients who visited the ED of Singapore General Hospital (SGH) with about one in every five Singaporeans aged 60 years or older.
- Every year, the SGH ED receives more than 120,000 visits and refers over 36,000 patients for inpatient admissions. EHR data analyzed in this study were obtained from Singapore Health Services.

Table 1. Description of various methods

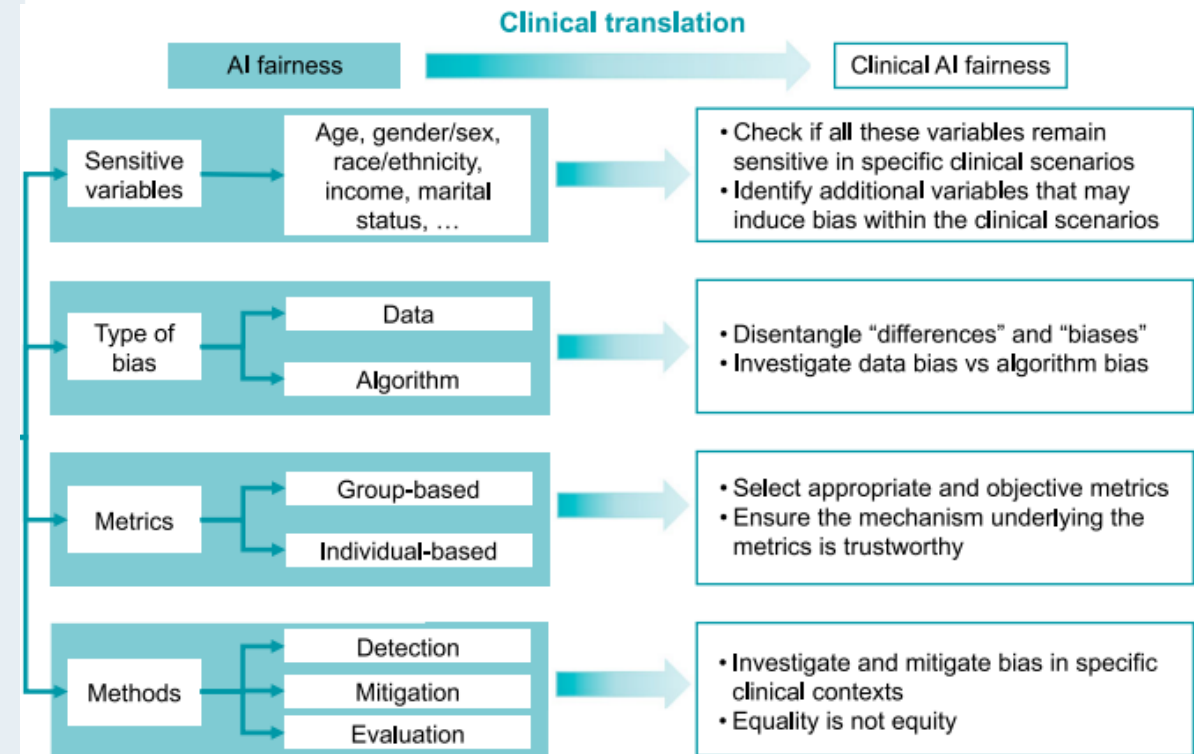
Classification	Models	Proportional hazards Assumption	Interpretability	Parameter tuning
Traditional statistical method	CoxPH model [5]	Yes	High	No
	Stepwise CoxPH [32, 33]	Yes	High	No
	CoxEN [34, 35]	Yes	High	No
Ensemble machine learning	RSF [6]	No	Moderate	Yes
	GBM [36]	No	Moderate	Yes
Interpretability machine learning	AutoScore-Survival [10]	Yes	High	Yes
Feedforward deep neural network	DeepSurv [7]	Yes	Low	Yes
	CoxTime [37]	No	Low	Yes
	DeepHit [38]	No	Low	Yes



Misuse of ML and AI Fairness

Box 1 | Recommendations to avoid overuse and misuse of AI in clinical research

1. Whenever appropriate, (predefined) sensitivity analyses using traditional statistical models should be presented alongside ML models.
2. Protocols should be published and peer reviewed whenever possible, and the choice of model should be stated and substantiated.
3. All model performance parameters should be disclosed and, ideally, the dataset and analysis script should be made public.
4. Publications using ML algorithms should be accompanied by disclaimers about their decision-making process, and their conclusions should be carefully formulated.
5. Researchers should commit to developing interpretable and transparent ML algorithms that can be subjected to checks and balances.
6. Datasets should be inspected for sources of bias and necessary steps taken to address biases.
7. The type of ML technique used should be chosen taking into account the type, size and dimensionality of the available dataset.
8. ML techniques should be avoided when dealing with very small, but readily available, convenience clinical datasets.
9. Clinician–researchers should aim to procure and utilize large, harmonized multicenter or international datasets with high-resolution data, if feasible.
10. A guideline on the choice of statistical approach, whether ML or traditional statistical techniques, would aid clinical researchers and highlight proper choices.

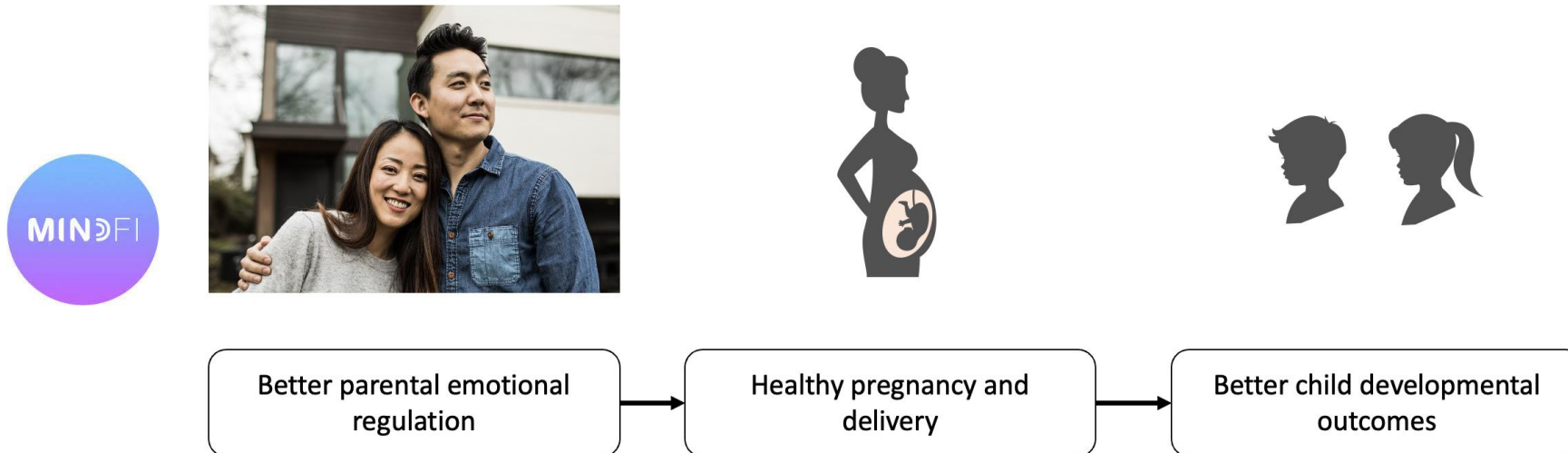


Liu, Mingxuan, et al. "A translational perspective towards clinical AI fairness." *NPJ Digital Medicine* 6.1 (2023): 172.

Volovici, Victor, et al. "Steps to avoid overuse and misuse of machine learning in clinical research." *Nature Medicine* 28.10 (2022): 1996-1999.

Ongoing Work

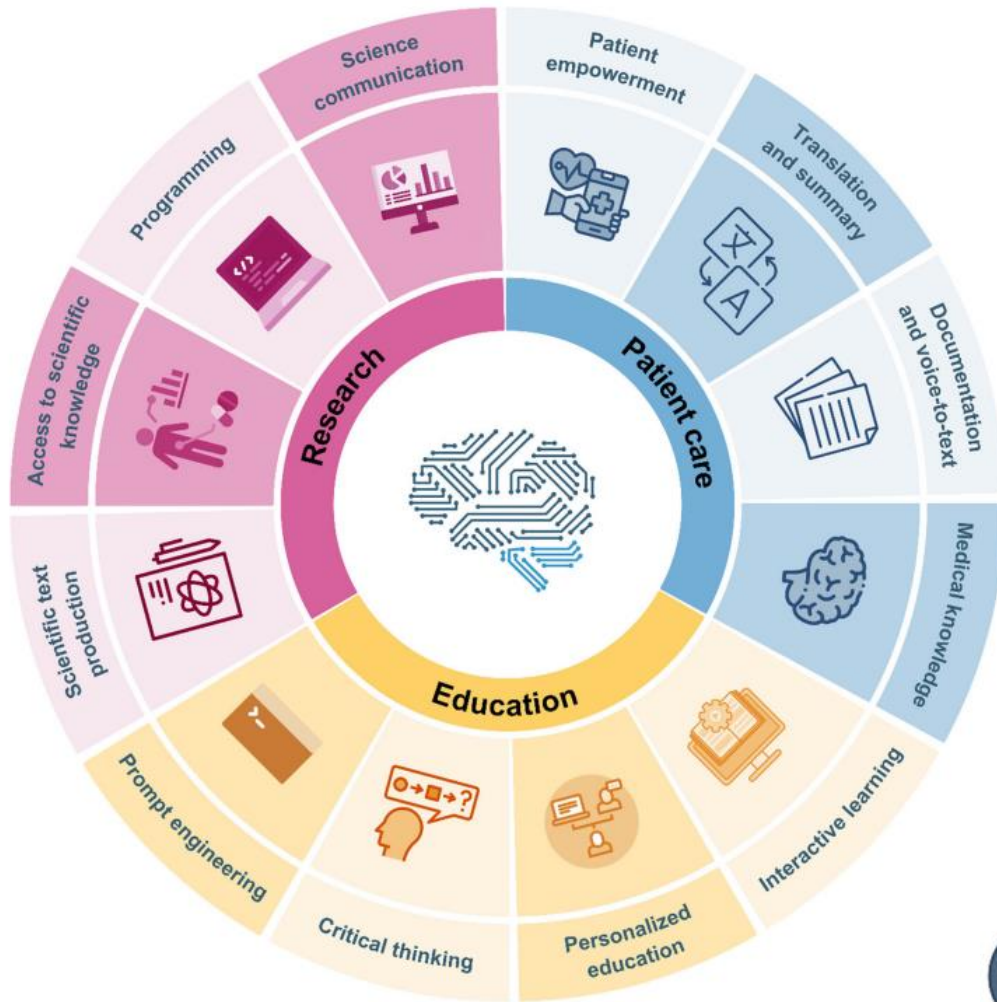
- Prenatal emotional regulation is key to postnatal parent-child relationship and child development.
- Studies have shown that the **antenatal** maternal health is a primary influence for child developmental outcomes.
- **Early interventions for parental emotional well-being** (both moms and dads) during the **antenatal** period can **improve parenting and the child's cognitive and emotional outcomes**.
- An mHealth study led by the A*STAR Singapore Institute for Clinical Sciences - Human Development Research Centre. At design phase.
- **Overarching goal:** **Implement, develop and evaluate** the effectiveness of an **adaptive, digital intervention**, through a mental wellness mobile application, on helping **expectant couples regulate their emotions and eventually improve parenting efficacy**.



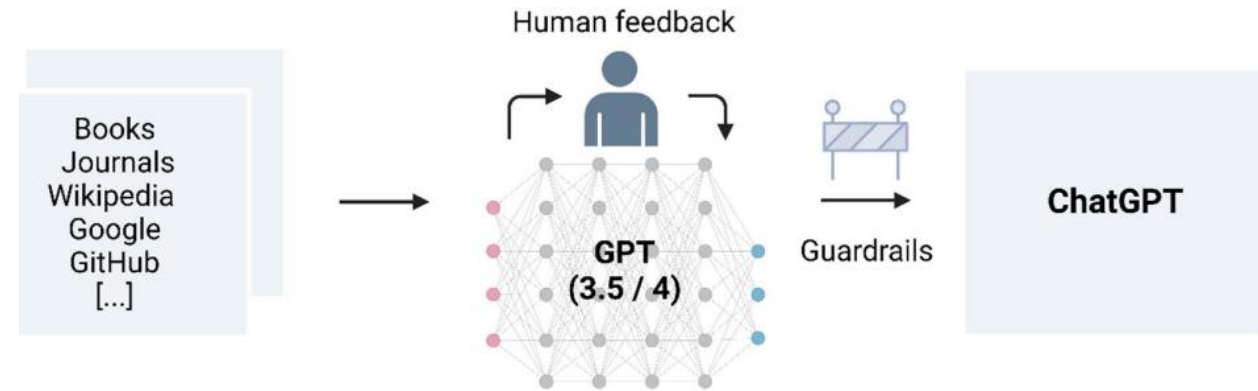
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The future landscape of AI in Health



- Simplified design of the architecture behind ChatGPT, including training, iterations of reinforcement learning by human feedback, choice of available model and implementation of guardrails to improve safety.



- Overview of potential applications for LLMs in medicine, including patient care, research, and education.



Static knowledge

Bias reproduction

Misinformation

Shortage in knowledge
and skills

Lack of
accountability



- Limitations of LLMs in their current state